

# Traffic Scheduling Simulation and Assignment for Area-Wide Evacuation

Yi-Chang Chiu

**Abstract**—Large-scale evacuation is called for when a natural or man-made extreme event strikes a populated area so that the population is exposed to immediate or foreseeable life-threatening danger. Evacuating a large population is an extremely complicated and difficult task, which primarily relies on efficient utilization of transportation systems, and effective evacuation schemes. This paper presents research that is aimed at developing an emergency evacuation modeling capability, which entails a careful integration of an optimal evacuation time and route choice model, as well as a traffic simulation model. Experiment results demonstrate the proposed methodology's operational planning capability for large-scale emergency evacuation management.

## I. INTRODUCTION

Large-scale evacuation is called for when a natural or man-made extreme event (e.g. hurricane, hazmat release, or dirty bomb) strikes a populated area so that the population is exposed to immediate or foreseeable life-threatening danger. Evacuating a large population is an extremely complicated and difficult task, which primarily relies on efficient utilization of roadway capacities and Intelligent Transportation Systems (ITS) technologies and modeling techniques, and effective and coordinated evacuation schemes. For events with a high probability of occurrence and sufficient lead time, such as a hurricane, most of the emergency management agencies determine alternate evacuation routes a priori. When a hurricane strikes, citizens are given advanced advisory or mandatory warnings to evacuate to safe neighboring cities by using those pre-determined evacuation routes. However, in an unexpected extreme event scenario, evacuating a large population could be more devastating due to the short response time. Without proper coordination in choosing evacuation times and routes, evacuees often get stuck in the roadway gridlocks for an excessive period of time. This could lead to significant life and property loss that may have been otherwise saved.

In the past, emergency evacuation schemes have been

studied for various disaster strikes such as earthquakes, tornadoes, fires, and hurricanes. These evacuation procedures were developed using different method such as the following:

A study conducted by Pal, Graettinger et al. [1] investigated emergency evacuation modeling using a Geographic Information System (GIS), with a focus on the coast of Alabama. Murray and Mahmassani [2] displayed the effectiveness of a simulation assignment tool DYNASMART-P in conjunction with two linear integer programs. This study created an evacuation-modeling framework that incorporates the trip-chain effect in times of evacuation. The proposed method was tested only on a small hypothetical network. Cova and Johnson [3] presented a network flow model for identifying optimal lane-based evacuation routing plans in a complex road network. The purpose of this research was to reduce evacuation delay times on street intersections. The case study for the lane-based evacuation routing was tested on the Salt Lake City, Utah area network.

Wilmont and Mei [4], developed alternative forms of trip generation of evacuation traffic and compared them with customary evacuation demand models. This research conducted tests on the southwest part of Louisiana following Hurricane Andrew. With the use of the data set of evacuation behavior collected from Hurricane Andrew, logistic regression and different forms of neural network models were tested, and both were concluded to be effective flow prediction models.

Nakanishi, Kim et al. [5] developed performance indicators that measure the benefits of having a transit agency prepared for emergency situations. Each agency would then perform self-reviews, address alternatives to the agency's vulnerabilities, and develop a cost benefit analysis to accommodate the selection of the optimal alternatives. These methods would then prepare transit agencies for emergency situations whether on or off the transit routes. Another hurricane evacuation related study proposed by Barrett, Ran et al. [6] studied the development of a dynamic hurricane evacuation-modeling framework for the use of long-term and short-term planning, in addition to real time operational purposes. It focused more on the programming and planning aspect of emergency management than in

Manuscript received 7/28/2004. Prof. Yi-Chang Chiu is with the Department of Civil Engineering, University of Texas at El Paso, El Paso, TX 79968 (phone: 915-747-6918; fax: 915-747-8037; e-mail: [chiu@utep.edu](mailto:chiu@utep.edu)).

evacuation operations.

Computer simulation models for network evacuation have been utilized for almost two decades. A research study by Radwan, Hbeika et al. [7] ran a macroscopic computer simulation model for an evacuation from highway networks undergoing a crisis. The model was tested on a Virginia area network. This study proved that the use of traffic simulation is an effective and efficient method to determine evacuation routes under disastrous conditions. Farahmand [8] developed a simulation model to prognosticate the optimal evacuation routes from the coastal areas of the Rio Grand Valley. With the use of Witness 7.0, a graphical interface software package, the network model was developed for evacuation for the area. Simulation tests concluded that the use of unidirectional traffic management is the best method for studying large evacuation participation rate.

In summary, most of the prior research in the literature focuses on varying evacuation scenarios. The common issues related to all above research are the capability to realistically describe the traffic evolution in time of emergency, and to develop a computationally effective algorithm to solve for the optimal evacuation schemes for a city or regional network. Few of the studies address the evacuation scheduling in conjunction with route choices and assignments. The primary goal of this research is therefore to improve the modeling capabilities in these areas.

## II. RESEARCH METHODOLOGY

Improving the current state-of-the-practice of emergency evacuation requires a fundamental breakthrough in modeling techniques. The theoretical challenges are two fold:

- First, it requires constructing an optimization model. This model minimizes casualties, exposure or other relevant measures by determining the optimal evacuation schedules of different zones of the area and determining the optimal routes in conjunction with the evacuation schedule.
- Second, it requires accurate estimations of traffic conditions based on traffic loadings resulting from varying evacuation time and route scenarios.

The present research can be understood with Figure 1. In this framework, the integration of an optimal evacuation-scheduling algorithm, a traffic simulation model (an enhanced version of DYNASMART-P with additional evacuation modeling features) and a Minimal Exposure Traffic Assignment (META) algorithms is necessary. The META algorithm includes the Minimal Exposure Shortest Path Algorithm and the traffic assignment procedure. At the convergence of the overall algorithm, the sought solutions include the optimal zonal evacuation departure

time schedule from evacuation zones to safe zones in conjunction with the corresponding optimal dynamic traffic assignment under the objectives of minimizing evacuation time and exposure.

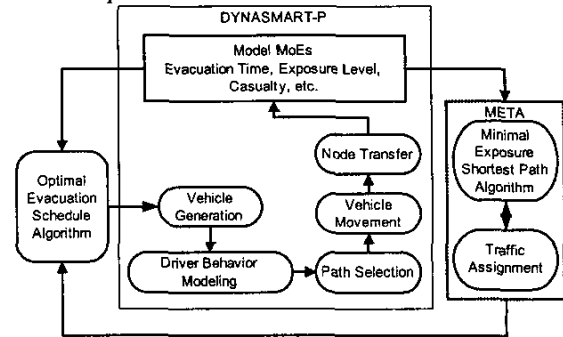


Figure 1. Minimal Exposure Traffic Assignment (META) Conceptual Structure

### A. DYNASMART-P

Dynamic Network Assignment Simulation Model for Advanced Road Telematics for Planning applications, also known as DYNASMART-P, is a state-of-the-art dynamic network analysis tool that was originally developed at the University of Texas at Austin (Jayakrishnan, Mahmassani et al. [9]; Chiu and Mahmassani et al. [10]; Mahmassani, et al. [11]) and has continued its development at the University of Maryland at College Park (Chiu and Mahmassani [12]; Mahmassani, Sbaiti et al. [13]). DYNASMART-P is recognized as an effective tool in areas such as intelligent network designing, planning, evaluation, and traffic simulation.

DYNASMART-P models individual travelers traveling adjustments as well as representing travelers pursuing to achieve a connection of activities at different points/locations in a network. Several known restrictions of static tools used in prevalent practices are overcome by DYNASMART-P. Details of DYNASMART-P can be found in (Chiu and Mahmassani [12]; Mahmassani, Sbaiti et al. [13]). For the purposes of this study, features that are critical for the evacuation modeling have been developed and incorporated into DYNASMART-P. These features include:

- **Zonal aggregation for minimal-exposure flow modeling**

The traffic generation in DYNASMART-P is based on the specified zonal demand during each demand generation interval. Vehicles are generated on the links, specified as "generation links", based on link capacity of user-specified weights. For evacuation modeling, safe zones and model inputs may be designated, but the traffic assignment to destinations in safe zones cannot be pre-specified, and have to be solved by the model. To address this issue, in the present research, all the designated safe zones are

aggregated into to a super safe zone. The super safe zone has a artificial centroid, which is modeled to be the “super sink node” for all the evacuation flows. The flows assigned to the safe zones are collapsed into one single zone. All flows are destined at the super sink node, by way of the physically exist destinations in the safe zones. The advantage of such a zonal aggregation is that more than one Traffic Analyze Zone (TAZ) can be designated as a safe zone. Once these safe zones are aggregated to a super zone, the model will automatically determine optimal traffic assignment to all the destinations nodes in theses safe zones. Figure 2 shows an example in which nodes 6,7, and 8 are the destinations. Their associated zones are specified as safe zones that are aggregated to super zone with centroid node 9. The connectors are artificial links with very high speed and infinite capacity. Traffic statistics on these connectors are not included in the final statistics.

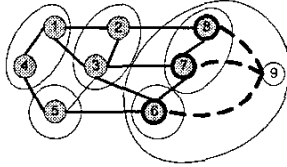


Figure 2 Zonal Aggregation for minimal-exposure flow modeling

#### • Improved mesoscopic traffic flow model

In the simulation core of DYNASMART-P, the vehicle movement during each time step is based on the speed in the link resulting from the density at the end of the previous time step. All moving vehicles in a link move at the same speed during each time step, depending on the density. The speed-density relationship currently used is a modified version of the Greenshield's equation. A specified minimum speed at jam density ensures that the simulation does not 'shut down' due to zero speeds. The relationship used is:

$$v = v_0 + (v_f - v_0) \left(1 - k/k_j\right)^\alpha$$

where,

$v_0$  = a user-specified minimum (jam) speed,

$v_f$  = free-flow speed of the highway segment,

$k$  = link density,

$k_j$  = density at the jam speed, and

$\alpha$  = a user-specified parameter.

This speed-density based flow model has shown to obey flow conservation [9] and is advantageous for large network application because it does not require the partitioning of long links into short ones, thus making it more computational efficient. However, the drawback is that it assumes a fast forward-moving shockwave exists

(leading vehicles are affected by trailing vehicles in each clock tick, even the trailing vehicles are miles behind). Second it does not explicitly handle flow discontinuities resulted from major capacity reduction or restorations. It often shows prolonged congestion and delay under certain traffic conditions. The present study overcomes this issue by combining a flow-density-based simulation procedure. Under normal traffic conditions, a density profile for each link is evaluated. For modeling flow discontinuities, a flow-density consistent with the designated speed-density function is used. Three flow states are explicitly considered for a link when major flow discontinuities take place. Vehicles movements are then updated based on the estimated shockwave speed and flow states propagation. Details of this procedure are documented in [14].

#### B. Evacuation departure time scheduling and traffic assignment

An effective evacuation scheme should consider both evacuation departure time scheduling, determination of safe zone(s), and traffic assignment. Evacuation departure time scheduling concerns the sequence of evacuation zones to be evacuated. This decision will have to be coupled with traffic assignment decisions because one affects the other.

In the present research, evacuation scheduling is formulated as a mathematical programming model: Given  $r_i$ , the objective is to minimize  $z = \sum_{i,k,\tau} r_{i,k}^\tau \cdot e_{i,k}^\tau$  by making  $r_{i,k}^\tau$  decisions, subject to (1)  $r_i^\tau = \sum_k r_{i,k}^\tau$ ,  $\forall i \in \text{evacuation zones}$ ,  $\tau \in \text{feasible departure time interval}$ ; (2)  $r_i = \sum_\tau r_i^\tau$ ,  $\forall i \in \text{evacuation zones}$ ; and (4) other traffic dynamic constraints.

The zonal aggregation as mentioned previously allows the flows to be destined to one zone regardless the number of original TAZs that are designated as safe zones. The major challenge of this problem concerns the departure time dimension. The decision variable  $r_{i,k}^\tau$  is found through a set of heuristic algorithms. Details of this problem formulation and solution procedure are documented in [15].

#### C. Experiment Test Bed Data

The methodology was tested on the City of El Paso, TX network that consists of 2909 nodes, 8295 links and 681 TAZs. The demand data includes aggregated personal and truck trips combined all trip purposes for both internal and external trips. The network was originally received in TransCAD format from the El Paso MPO, and was later converted to DYNASMART-P format. An image of the network shown in DYNASMART-P is illustrated in Figure 3.

#### D. Plume Dispersion Scenario

A hypothetical event was developed to test the proposed evacuation routing algorithm. A local refinery, Western Refining Company, was chosen as a location of a possible event due to its central location within the city of El Paso. The refinery is approximately three miles east of the downtown central business district and is embedded within a residential area and near several active shopping centers.

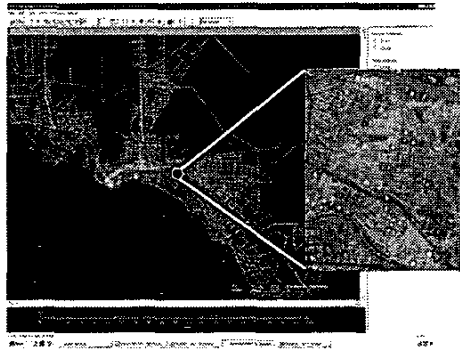


Figure 3. Western Refining Company Location

The scenario consists of an extreme event occurring within the refinery, creating a plume of a hazardous material. Because the objective of this research is to demonstrate and examine the proposed evacuation routing algorithms, reasonable assumptions have been made on the dispersion of the hazardous gas. The plume dispersion patterns can be further improved by applying a meteorological plume dispersion model. With toxic and life threatening airborne hazmat dispersing throughout central El Paso, and based on historical meteorological data, the temporal and spatial progression of the airborne hazardous material dispersion was estimated to travel 2 to 3 miles within 30 minutes, 4 to 5 miles within 45 minutes, and 6 to 7 miles within 60 minutes after the incident occurrence. Figure 4 indicates visual images of the different time frames with the hazmat plume being illustrated in red-color marks. Reading from left to right the images display dispersion rates from 30, 45, and 60 minutes after incident occurrence.

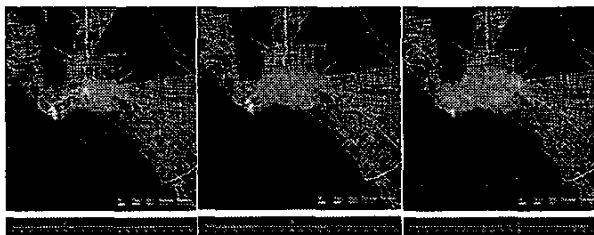


Figure 4. Plume Dispersion Scenario on El Paso network (from Left to Right, 30, 45, 60 Minutes After the Start of Simulation)

### III. EXPERIMENT DESIGN

The experiments conducted in this study are briefly described as follows:

#### A. Base Case (BC)

The (BC) scenario represents the traffic flow pattern of tripmakers under a normal traffic condition. Taking the input travel demand data, DYNASMART-P determines the time-dependent user-equilibrium (UE) paths and the corresponding traffic assignment onto the network. Under this scenario, the demand pattern is that of the morning peak-hour traffic, with heavier directional traffic heading toward the Central Business District (CBD). In total, about 280,000 vehicles are generated and simulated throughout all the scenarios in the 100-minute simulation period.

#### B. Hazmat without information (HWOI)

This scenario is to investigate the impact of the hazmat plume under the same UE assignment in terms of numbers of tripmakers who are under the exposure of the hazmat. The temporal dispersion of the plumes follows the scheme described in the preceding section. It is assumed that 60 minutes after the initial start of the hazmat release, the entire central El Paso area is exposed to the hazmat (Figure 4). For benchmarking purpose, it is also assumed that all the trip makers do not have knowledge about the hazmat release. As such, trip makers continue trips to the originally intended destinations. If the intended destinations are within the exposure zones, then the tripmakers are exposed to the hazmat.

#### C. Hazmat with descriptive information (HWDI)

This scenario examines the effectiveness of descriptive information for mitigating the hazmat release. In this scenario, it is assumed that the hazmat release information is broadcasted to the public through various media (TV, radio, website, etc.). Tripmakers are aware of the hazmat dispersion pattern at the time of departure to their destination, and try to find a minimal-risk route. In this scenario, tripmakers, however, do not evacuate to shelters and still intend to arrive to their original destinations. It is also assumed that once the tripmakers get in to the vehicles they do not have further information regarding to the spatial dispersion of the hazmat. The hazmat plume dispersion scenario is the same as that in the HWOI scenario.

#### D. Hazmat with Minimal-Exposure Traffic Assignment (HMETA)

This scenario examines the effectiveness of Minimal-Exposure Traffic Assignment (META) proposed by this research. The META is applied to provide the minimal exposure routing to individual tripmakers who may be potentially exposed to the hazmat. In this scenario, the

original origin-destination trip demand is not disturbed. Therefore, it is assumed that those who go in and out of the hazmat exposure zones will be subjected to the exposure; those who traverse through the exposure area will be re-routed to avoid the hazmat exposure.

#### E. Hazmat with Evacuation and Minimal-Exposure Traffic Assignment (HEMETA)

This scenario further applies the evacuation scheme in addition to the META. The evacuation scheme advises those tripmakers who are in the exposure zones to evacuate to the nearest evacuation zones in a timely manner. For those whose original destinations are in the contaminated zones, they are asked to postpone or cancel the trips. In this case, not only do the trip makers need to be evacuated before the hazmat plumes arrive, but also need to be assigned to the routes that allow for speedy exit of the exposure zones without being exposed to the hazmat. In this scenario, both the optimal evacuation shelter assignment and the META problems are solved jointly. In this scenario, the citizens are assumed to be in full compliance of the evacuation advisories, which is a quite optimistic assumption. Additional discussions of the impact of realistic evacuation advisory compliance behavior can be found in Chiu, Mirchandani, Korada [16].

#### IV. EXPERIMENT RESULTS

As shown in Figure 5, the average travel times for all the tripmakers in the (BC) scenario is 11.2 minutes. If the hazmat release occurs and no mitigation action is taken (HWOI), the number of vehicles exposed to the hazardous materials is more than 107,600, approximately one third of the total generated vehicles in the simulation period. If the incident information is provided to the public (HWDI), the number of exposed vehicles are reduced to approximately 68,000 vehicles.

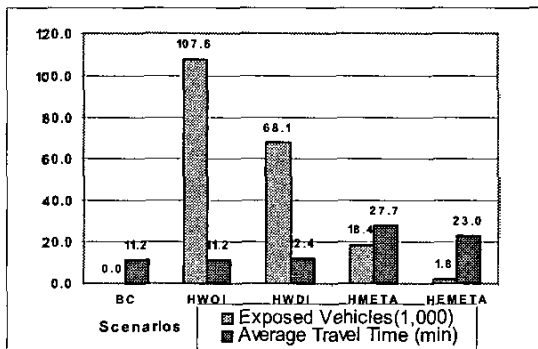


Figure 5. Evacuation Scenario Comparison

Although significantly less than that in the HWDI scenario, this number is still considered excessive. Nonetheless, the reduction of exposed vehicles from HWOI to HWDI scenario is attributed to the incident information

received by the public so that trip makers are aware of the prevailing temporal and spatial dispersion progressions of the hazmat at the time of departure. However while en-route to the destination, tripmakers do not have further updates on the hazmat, thus may take a route that is contaminated. Furthermore, because no evacuation is put in effect, those whose final destinations are in the contaminated zones are eventually exposed to the hazmat. In the HWDI scenario, the average travel time increase 12.4 minutes from the 11.2 minutes in the BC scenario. It is primarily because the re-routing to longer routes due to the incident.

If the Minimal-Exposure Traffic Assignment (META) is performed, the number of exposed vehicles is further reduced to about 18,400. META allows trip makers to collectively find the minimal-exposure routes to their destinations. There is a trade-off between the exposure reduction with the increased average travel time because more vehicles are routed through longer paths to arrive at the destinations. An example of such a diversion is illustrated in Figure 6. The minimal-exposure path from the east to the west side of El Paso is determined to be Loop 375 instead of the IH-10, since the downtown section has been entirely exposed to the hazmat. This type of assignment significantly reduced citizens' risk of exposures by re-routing them away from the contaminated areas and using other portions of the highway networks that could have been otherwise underutilized.

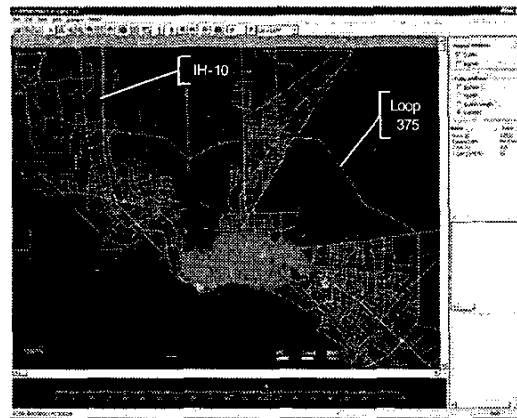
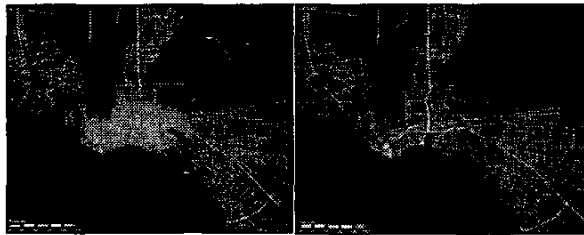


Figure 6. Minimal-Exposure Path from East El Paso to West El Paso

Furthermore, if the META in conjunction with the evacuation scheme is implemented, the number of the exposed vehicles is further reduced to less than 2,000. This result is mainly attributed to the timely evacuation of those citizens in the contaminated zones, which is accomplished through the forecast and simulation of the hazmat dispersion, and the minimal-exposure routing and evacuation in conjunction with the evacuation scheduling. In this scenario, about 15,000 vehicles are evacuated out of

the contaminated zones. Moreover, the travel time also improves comparing with the HMETA scenario. A likely reason for this occurrence is that about 23,000 vehicles cancel their trips into the contaminated zones, and result in a smaller number of vehicles in the network.



**Figure 7. Network Traffic Flow Comparison between BC and HMETA scenarios**

The temporal and spatial variation of plume dispersion and traffic dynamics can be visualized in DYNASMART-P. Figure 7 illustrates the traffic flow throughout the network between the BC and HMETA scenarios. In this diagram, network links are color-coded based on the link traffic density (number of vehicles per traffic lane per mile). Higher density indicates greater congestion. As shown in the diagram, scattered congestion in the east El Paso area is observed, which is consistent with morning traffic patterns in El Paso. Having the incident in place and HMETA implemented, the traffic flow pattern on the network has been significantly altered. Obvious congestion is observed on more urban streets in the northeast, north side (south of El Paso International Airport, and Fort Bliss), and Sunland Park areas on the west side of El Paso.

#### V. CONCLUDING REMARKS

This paper presents research that is aimed at development of an emergency evacuation model through a careful integration of an optimal evacuation time and route choice model, as well as a traffic simulation model. The El Paso area network was established as a test bed for the case study in which a toxic airborne hazardous material release incident occurs in the morning during peak traffic conditions. The computational experiments show that the proposed methodology provides a critical operational planning capability for large-scale emergency evacuation management in response to an unexpected extreme event. Significant insights about evacuating citizens out of central El Paso were provided.

#### VI. ACKNOWLEDGEMENT

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